

- 5 (a) Fig. 5.1 shows two small loudspeakers, L_1 and L_2 , separated by 15 cm. A microphone is moved along a line XY parallel to the line joining the two loudspeakers and at a perpendicular distance of 1.2 m away.

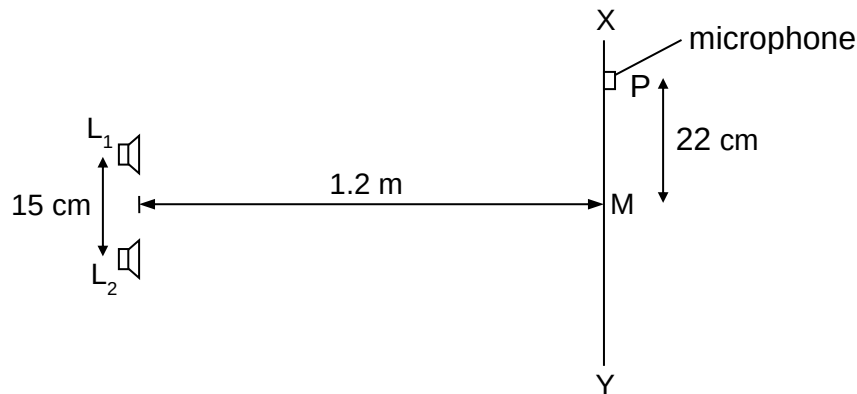


Fig. 5.1

The centre of the interference pattern formed along XY is at point M. When the microphone is moved from M to P by a distance of 22 cm, it detects three intensity maxima including the ones at M and P.

Given that the speed of sound in air is 330 m s^{-1} , determine the approximate frequency at which the speakers were driven. Express your answer to 2 significant figures.

frequency = Hz [3]

Method 1

three intensity maxima detected from M to P including the ones at M and P

→ 2 fringe separations = 22 cm

→ fringe separation, $x = 11 \text{ cm}$

Two-source interference fringe separation,

$$x = \frac{\lambda D}{a}$$

$$11 \times 10^{-2} = \frac{\lambda(1.2)}{15 \times 10^{-2}}$$

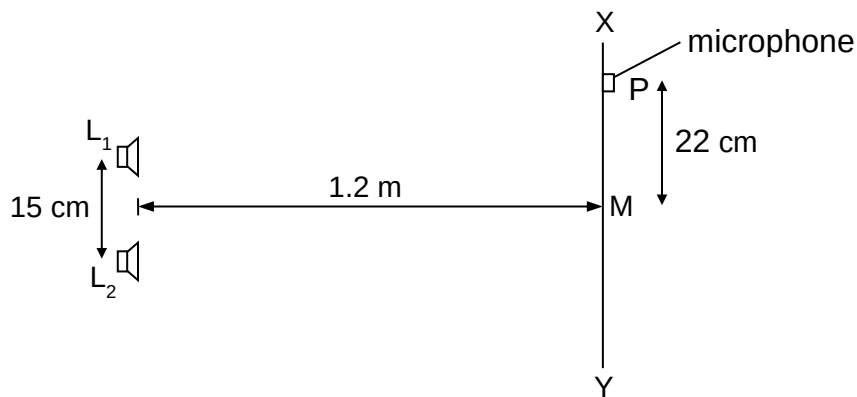
$$\therefore \lambda = 0.01375 \text{ m}$$

$$\therefore f = \frac{v}{\lambda} = \frac{330}{0.01375} = 2.4 \times 10^4 \text{ Hz}$$

1

1

1

Method 2

Finding the path difference,

$$\text{Path of } L_1P = \sqrt{1.2^2 + 0.22 - 0.075^2} = 1.209 \text{ m}$$

$$\text{Path of } L_2P = \sqrt{1.2^2 + 0.22 + 0.075^2} = 1.236 \text{ m}$$

$$\text{Path difference} = 1.236 - 1.209 = 0.027 \text{ m} = 2\lambda$$

$\therefore \lambda = 0.0135 \text{ m}$ (Note that it is a different outcome from the Young Double Slit experiment. It is expected as the Young's Double Slit equation is an approximation to the fringe separation.)

$$\therefore f = \frac{v}{\lambda} = \frac{330}{0.0135} = 2.4 \times 10^4 \text{ Hz}$$

- (b) Fig. 5.2 shows a microwave transmitter T and a microwave receiver R placed at the same angle θ to the normal of a horizontal board A, which partially reflects and transmits microwaves. A similar horizontal board B is placed a distance d below board A, such that a high intensity signal is detected by receiver R. A metal sheet is placed between T and R to prevent microwaves from reaching R directly from T.

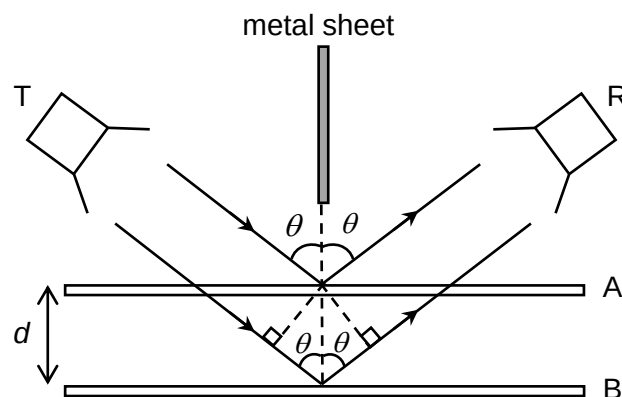


Fig. 5.2

The path difference between the two waves reflected off boards A and B is given by

$$2d \cos \theta$$

When a high intensity signal is detected by R,

		$2d \cos \theta = m\lambda$ where m is a positive integer and refers to the order of constructive interference ($m = 1, 2, 3, \dots$) and λ is the wavelength of the microwaves.	
	(i)	State the phase difference, in radians, between the reflected microwaves from A and B at a point where a high intensity signal is detected.	
		phase difference = rad	[1]
		0 rad (or 2π rad, 4π rad, etc.), as a high intensity signal or constructive interference is detected.	1
	(ii)	When distance d is increased by lowering board B, alternating low and high intensity signals are detected by receiver R. Explain these observations.	
			
			
			[3]
		As board B is lowered, the path difference between the two reflected waves continuously increases .	1
		This causes the phase difference between the two waves at R to continuously alternate between being in phase and antiphase .	1
		The high intensity signals are due to constructive interference between the reflected waves whenever they meet in phase at R . The low intensity signals are due to destructive interference between the reflected waves whenever they meet in antiphase at R .	1
	(iii)	Transmitter T and receiver R are now placed side-by-side and facing the boards normally, meaning that $\theta = 0^\circ$. As board B is moved 140 mm downwards at a constant speed, receiver R goes from the initial high intensity signal through nine high intensity signals and then to a final high intensity signal. Determine the wavelength of these microwaves.	
		wavelength = m	[2]
		Since $\theta = 0^\circ$, the expression for path difference in terms of d now becomes: $2d \cos 0^\circ = 2d$ Distance moved by board B (d) $\times 2$ = Change in path difference = 10λ Thus, $2d = 10\lambda$ $140 \text{ mm} \times 2 = 10\lambda$ $\lambda = 28 \text{ mm}$ $= 0.028 \text{ m}$	1 1

6 In a photoelectric experiment, an ultraviolet (UV) light source of constant intensity and single